



AISSMS
COLLEGE OF ENGINEERING

ज्ञानम् सकलजनहिताय
Accredited by NAAC with "A+" Grade



**PRODUCTION ENGINEERING DEPARTMENT
(SANDWICH PATTERN)**

**DIGITAL PRODUCTION & TOOLS MONITORING SYSTEM FOR
FORGING SHOPS**

SUBMITTED BY

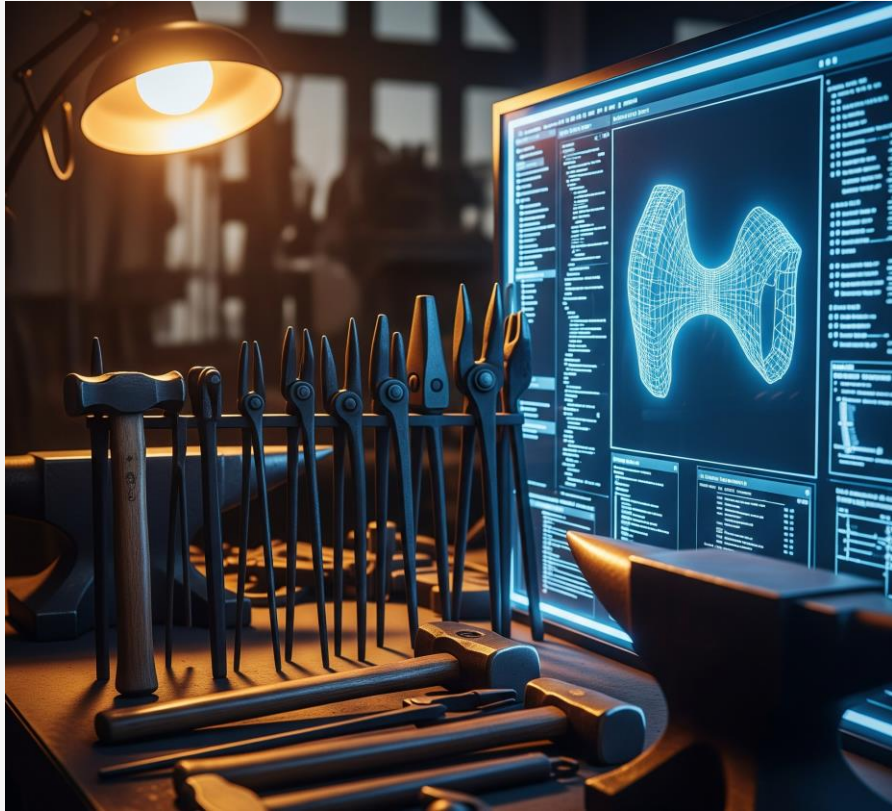
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EXAM SEAT NO: B400210601

UNDER GUIDANCE OF

MR.M.A KELKAR

INTRODUCTION



- **Digital Transformation** of legacy paper-based systems
- **Real-Time Task Reporting** for workers and supervisors
- **Digital Attendance** tracking with timestamps
- **Tool Management** system to track usage and lifecycle
- **Data-Driven Decisions** powered by exportable reports

Problem Statement

Before (Manual System)



Worker confusion, unclear instructions

Delayed reports, missed entries

✗ No traceability of tools/tasks

Paper gets damaged or lost

After (ForgeTrack System)



Easy-to-use digital interface

Real-time task updates

📁 Auto-recorded, searchable data

🔒 Secure, exportable reports

Scope

From small shops to
large-scale industries

ForgeTrack fits all

Manufacturing

**Construction
Equipment**

**Machining
Shops**

**Forging &
Fabrication**

Automotive

Aerospace

Wireframe Design

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ForgeTrack's interface was wireframed to ensure:

- Easy navigation for shopfloor workers
- Minimal text input to reduce reporting time
- Visual clarity for supervisors (color-coded tasks)
- Modular design to allow future upgrades (QR, mobile app)

- [Aditya's Forging Shop Digital Tracker Code Preview.one](#)
- [Aditya's Forging Shop Digital Tracker Code.png](#)
- [Aditya's Forging Shop Digital Tracker.png](#)
- [Aditya's Forging Shop Digital Tracker Preview.html](#)



ForgeTrack Interface – Dashboard & Task Reporting

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(Part 1)

- Home Page
- Task Reporting Form
- Layout: Side-by-side images with short captions

(Part 2)

- Attendance Module
- Tool Tracker
- Export to Excel

Features Shown:

- Navigation menu with core modules (Home, Report, Tools, Attendance)
- Task form includes worker name, machine, operation, quantity, and shift time
- Designed to minimize data entry time while maintaining report accuracy

Task Reporting

Worker ID

Task Type

Forging

Task Date

06/06/2025

Part Number

Tool ID Used

Comments

Excess Flash
Action Taken - Sent to trim again |

Submit Task

Daily Attendance Report

Worker ID

Date

06/06/2025

Status

Present

Submit Attendance

Aditya’s Forging Shop - Combined Report View

RecordType	WorkerID	TaskType	Date	PartNumber	ToolID	Comments	AttendanceStatus
Task	ADIT0310	forging	2025-06-06	T2024A	T2024A	Aluminum 400	
Task	ADIT0310	packaging	2025-05-24	T2024A	T2024A	Brass 1000	
Task	ADIT0310	machining	2025-07-01	T2024A	T2024A	Aluminum 2000	
Task	ADIT0310	forging	2025-06-06	FG2025-A01	T2024A	Excess Flash Action Taken - Sent to trim again	
Attendance	ADIT03102002		undefined				Present
Attendance	ADIT0310		2025-06-06				present
Attendance	ADIT0310		2025-05-24				present

Worker Details →

Attendance Module →

Tool Tracker → Export to Excel

This sequence ensures:

- Daily attendance is logged before task reporting
- Tool usage is mapped per shift
- All records are exportable — enabling review, Key Performance Indicators, and compliance

1. Attendance Module
2. Tool Tracker
3. Export to Excel

From shop floor to spreadsheet
ForgeTrack ensures traceability at every step

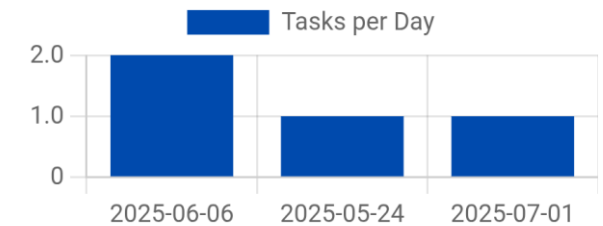
17 Generate Monthly Report

Select Month

June 2025

Generate Monthly Report

Daily Productivity Chart



17 Monthly Report for 2025-06

RecordType	WorkerID	TaskType	Date	PartNumber	ToolID	Comments	AttendanceStatus
Task	ADIT0310	forging	2025-06-06	T2024A	T2024A	Aluminum 400	
Task	ADIT0310	forging	2025-06-06	FG2025-A01	T2024A	Excess Flash Action Taken - Sent to trim again	
Attendance	ADIT0310		2025-06-06				present

Report & Analytics

 Bar Chart	Pie Chart	 Screenshot
 <i>Monthly Productivity</i> (e.g., tasks completed per worker or shift)	<i>Tool Life Status</i> (e.g., % Good, Worn, Needs Replacement)	<i>Excel Report Preview</i> (Actual exported file showing columns like Date, Task, Quantity)

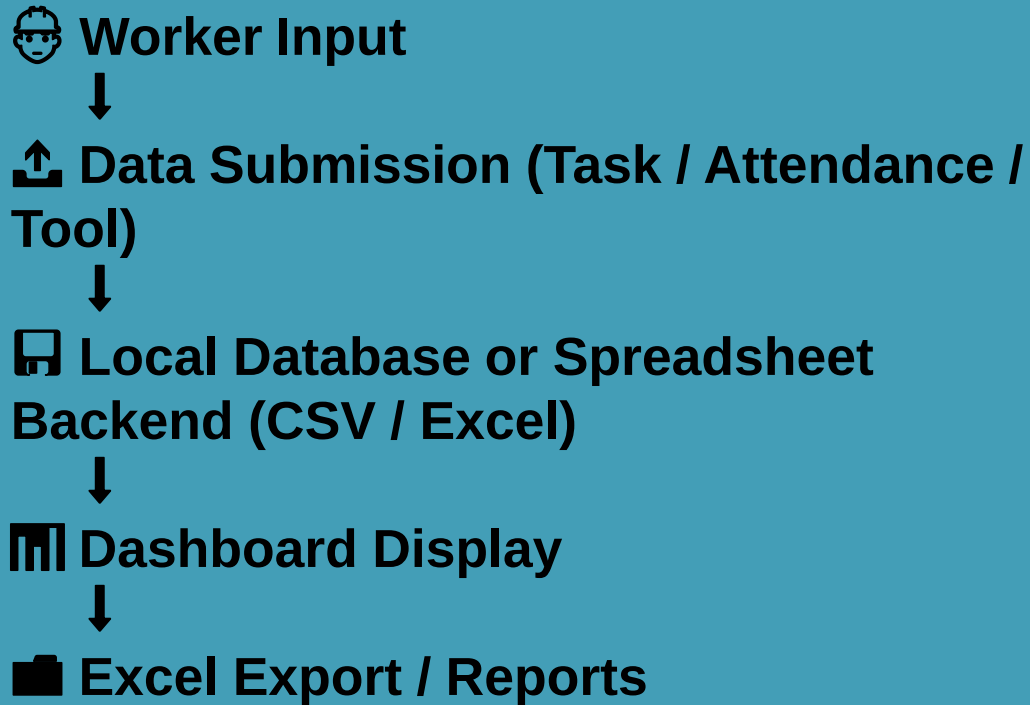
ForgeTrack auto-generates key reports such as:

- Task completion trends (daily, weekly, monthly)
- Tool usage statistics
- Attendance heatmaps (optional)
- Export-ready Excel files for audits or KPIs

These analytics help supervisors:

- ✓ Identify low performers
- ✓ Predict tool replacements
- ✓ Optimize shift allocations

System Architecture & Workflow



You can represent each block as:

User Input Form
Processing Module
Database/Storage Layer
Frontend Dashboard
Export / Analytics

Use arrows with labels like:

Submit Entry
Store Data
Display Records
Generate Report

The system uses a simple yet effective architecture:

- Frontend: Web-based form for easy inputs
- Backend: Local data stored in structured sheets (CSV/Excel)
- Processing: Data filtered and formatted for analysis
- Output: Charts, reports, and downloadable Excel summaries

Defined User Roles & Responsibilities

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-  **Worker**
- **Access:** Task reporting form, attendance input
- **Actions:**
 - Mark shift attendance
 - Fill daily task reports
 - View basic dashboard (optional)

-  **Supervisor**
- **Access:** Reports dashboard, tool tracker, attendance logs
- **Actions:**
 - Review daily task entries
 - Check tool life and usage
 - Monitor attendance logs
 - Export data to Excel

-  **Admin**
- **Access:** Full control panel
- **Actions:**
 - System settings
 - Add/remove users
 - Maintenance and updates
 - Backup/export data

Testing, Debugging & Real-World Validation

 Feature	 Tested?	 Result	 Bug Fixed?
Task Reporting	 Yes	Accurate entry logs	Minor input validation bug – fixed
Attendance Logging	 Yes	Time-stamped data	No bugs found
Tool Tracker	 Yes	Usage stats tracked	Fixed export glitch
Excel Export	 Yes	File generated cleanly	Header alignment fixed
Dashboard Display	 Yes	Graphs loaded correctly	No major bugs

Future Enhancements & Expansion Roadmap

My vision is to evolve ForgeTrack into:

- A cloud-enabled, mobile-first platform
- Fully integrated with smart attendance
- Capable of predictive maintenance & analytics

List of Future Enhancements

1. 📱 Mobile App Access

Operators can mark attendance or fill reports on mobile devices.

2. ☁ Cloud Integration

Data accessible across multiple locations securely via cloud.

3. 📷 QR Code-Based Attendance

Scan-and-go system for fast, contactless attendance.

4. 🔔 Smart Notifications & Alerts

Tool expiry alerts, missed attendance, daily task reminders.

5. 📊 AI-Based Productivity Insights

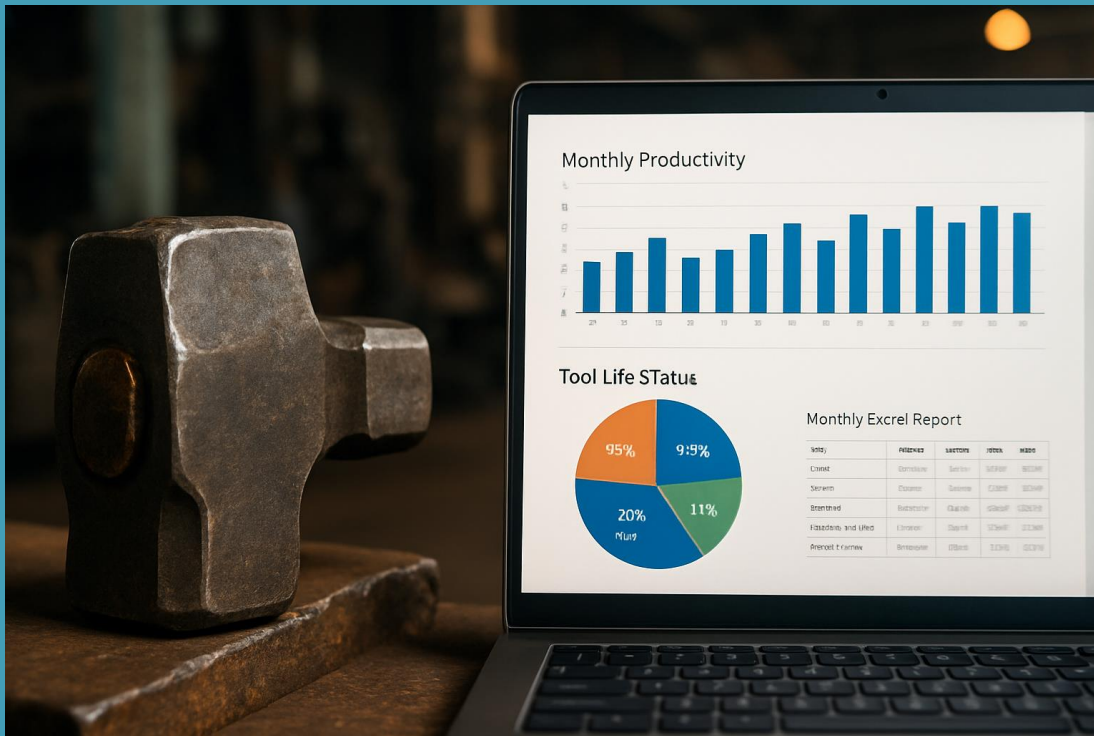
Auto-suggestions for shift planning or maintenance based on data patterns.

Challenges Faced & Solutions Applied

Every challenge was an opportunity to adapt, innovate, and improve the usability of ForgeTrack on the real shop floor.

⚠ Challenge	✓ Solution
Manual task tracking was inconsistent	Designed structured digital forms with dropdowns for standardization
Workers had low digital literacy	Created a minimal, icon-based UI with local language hints
Delay in attendance marking	Introduced timestamp logging and simplified the input screen
Tool usage wasn't being recorded	Built a basic tool tracker with lifecycle tagging
Excel exports were misaligned	Adjusted backend formatting to match header and data rows

Conclusion & Key Takeaways



1. What I Built:

ForgeTrack is a **modular, web-based task tracking and reporting system** designed to digitize operations on the forging shopfloor.

2. What It Solves:

- ✓ Eliminates manual errors
- ✓ Speeds up task & attendance logging
- ✓ Centralizes tool management
- ✓ Enables data-driven decisions

3. Key Outcomes:

- Increased transparency
- Real-time productivity tracking
- Foundation for smart manufacturing

References & Acknowledgements

- Websites & Documentation:

- MDN Web Docs – developer.mozilla.org
- W3Schools – w3schools.com
- GeeksforGeeks – geeksforgeeks.org
- Stack Overflow – stackoverflow.com

- Tools & Frameworks:

- Visual Studio Code
- Google Chrome DevTools
- Bootstrap
- HTML5, CSS3, JavaScript

- Charts & Analytics:

- Chart.js
- Microsoft Excel (for data export formatting)

THANK YOU!

Any Questions?

